

BAYESIAN HIERARCHICAL INFERENCE OF THE DARK MATTER HALO CONCENTRATION–MASS RELATION FROM MANGA DISK GALAXY ROTATION CURVES

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ABSTRACT

We present a three-stage Bayesian framework for inferring the dark matter halo concentration–mass (c – M) relation from MaNGA integral-field H α kinematics. An empirical rotation-curve screen first rejects galaxies whose velocity fields lack credible disk-like rotation; the survivors enter an NFW-based single-galaxy mass decomposition that yields joint (M_{200}, c) posteriors; and a prior-corrected importance-sampling likelihood propagates those uncompressed samples directly to a population-level c – M fit—bypassing any compression into means or covariance matrices. The final retained sample is 620 nearby late-type disk galaxies. The population fit yields $\log_{10} c_0 = 0.982^{+0.023}_{-0.024}$, $\alpha = -0.093^{+0.051}_{-0.053}$, and intrinsic scatter $\sigma_{\text{int}} = 0.237^{+0.027}_{-0.026}$ dex (95% credible intervals). The 95% ETI for α lies entirely negative, consistent with the Λ CDM late-assembly expectation for this kinematically selected subsample; it is, however, conditional on fixed photometric inclinations and broadens to encompass zero when a 5° Gaussian inclination prior is allowed. The primary contribution is this Bayesian workflow—propagating non-Gaussian, curved single-galaxy posteriors directly to population-level inference—rather than an observational calibration of the c – M relation.

Subject headings: dark matter halos — galaxy rotation curves — galaxy mass distribution — galaxy kinematics

1. INTRODUCTION

How dark matter halos build their internal structure—whether a halo’s concentration encodes its formation epoch or reflects subsequent baryonic modification—sits at the intersection of small-scale galaxy dynamics and large-scale cosmological assembly (S. D. M. White & M. J. Rees 1978; A. J. Benson 2010). In the standard Λ CDM picture, halos follow the NFW density profile (J. F. Navarro et al. 1997); its two parameters, virial mass M_{200} and concentration c , together specify the radial mass distribution from the inner cusp to the virial boundary. Massive halos form late: collapsing at lower background density, they settle into equilibrium with shallower inner potentials and suppressed concentrations—a trend reproduced across decades of cosmological simulation (R. H. Wechsler et al. 2002; J. F. Navarro et al. 2009). Both the normalization and slope of this anti-correlation shift with cosmology and redshift (A. A. Dutton & A. V. Macciò 2014; T. Ishiyama et al. 2021), so an empirical c – M measurement simultaneously constrains halo assembly history and the background cosmological model.

Rotation curves encode the total gravitational potential along the disk midplane (J. Binney & S. Tremaine 2008; S. Courteau 1997); their shapes vary systematically with morphology and stellar mass (Y. Yoon et al. 2021), making them a natural diagnostic of the dark-matter fraction across a range of radii. Integral-field spectroscopy replaces the single slit with a full

two-dimensional velocity map, exposing projection effects, signal-to-noise variation, and non-circular flows that long-slit extraction suppresses or averages away. MaNGA’s DAP delivers homogeneous H α velocity maps, pixel-level inverse-variance estimates, and structural parameters for $\sim 10\,000$ nearby galaxies—a ready-made platform for large-sample halo inference (K. B. Westfall et al. 2019). A velocity map, however, is not a mass model. Inclination uncertainty, beam smearing, asymmetric drift, bar-driven non-circular motions (E. O. Schmidt et al. 2023), and the classical disk–halo degeneracy all intervene between the projected line-of-sight velocities and the halo parameters inferred from them; translating the former into the latter without systematic bias is the core obstacle.

The problem has two distinct layers. At the single-galaxy level, the model must decompose rotational support into baryonic, NFW, and asymmetric-drift terms in a way that is physically interpretable yet cheap enough to run on hundreds of targets; at the population level, the inference must preserve the genuine uncertainty geometry of each galaxy’s posterior rather than substituting a low-dimensional surrogate. We anchor the halo-mass prior to the stellar-to-halo mass relation (SHMR) of B. P. Moster et al. (2013), whose applicability to disk-galaxy dynamics has been validated by recent rotation-curve analyses across a wide mass range (E. M. Di Teodoro et al. 2023; P. E. Mancera Piña et al. 2025). Bias is unavoidable when measurement-error geometry and intrinsic scatter are both neglected—a complication long recognized for astronomical regression (D. W. Hogg et al.

2010). It is especially severe here: single-galaxy (M_{200}, c) posteriors occupy narrow, curved ridges whose orientation encodes information about the slope α that is entirely lost the moment the posterior is compressed into a covariance matrix.

Our analysis follows a three-stage Bayesian pipeline. An empirical rotation-curve screen first rejects galaxies whose $H\alpha$ velocity fields are too disturbed for stable posterior sampling; the survivors then enter a full NFW-plus-baryonic-plus-drift mass decomposition, which draws posterior samples of M_{200} and c for each galaxy; and a population-level importance-sampling step propagates those raw samples to the c - M hyperparameters without intermediate compression. No means, covariances, or other low-dimensional summaries are computed between stages. The goal is a workflow in which non-Gaussian, correlated single-galaxy posteriors propagate faithfully to population-level inference—applied here to a strictly screened subsample of nearby MaNGA late-type disk galaxies and benchmarked against Λ CDM expectations.

Two questions frame the analysis. Can MaNGA two-dimensional gas kinematics, paired with NFW dynamical modeling, yield credibly constrained M_{200} and c for a sample of this size? And if so, do the propagated posterior samples recover a c - M slope consistent with Λ CDM expectations for late-type disk galaxies? Section 2 describes the data products, the empirical screening model, and the final sample definition. Section 3 presents the single-galaxy dynamical model, prior structure, and likelihood. Section 4 introduces the population c - M model and its importance-sampling likelihood. Section 5 presents and discusses the results. Section 6 summarizes.

2. DATA AND SAMPLE SELECTION

2.1. MaNGA Data Products

Deployed as one of the core programs of SDSS-IV, MaNGA (Mapping Nearby Galaxies at Apache Point Observatory) acquires spatially resolved spectra for $\sim 10\,000$ nearby galaxies via multiplexed IFU fiber bundles of varying angular size (K. Bundy et al. 2015). The two-dimensional coverage matters: each bundle delivers a velocity map across the full disk face, exposing inclination gradients, beam-smearing, and non-circular perturbations that a single slit would fold irreversibly into the extracted profile—making MaNGA the natural substrate for the velocity-field reconstruction, empirical screening, and dynamical decomposition carried out here.

All observational inputs derive from the final SDSS-IV MaNGA release (DR17; Abdurro’uf et al. 2022). From the DAP MAPS product (SDSS-IV Collaboration 2024a; K. B. Westfall et al. 2019) we extract the $H\alpha$ line-of-sight velocity field and its per-spaxel inverse variance, the bin-level SNR, elliptical polar coordinates (r, ϕ), photometric position angle and axis ratio, and the gas velocity-dispersion map; the DRPALL summary catalog (SDSS-IV Collaboration 2024b) supplies the redshift, photometric half-light radius R_e , stellar mass M_* , and Sérsic index n (Table 1).

2.2. Empirical Rotation-Curve Screening

Each velocity field enters the pipeline not directly into the physical mass model, but first through a lightweight

TABLE 1
MaNGA OBSERVATIONAL INPUTS TO THE DYNAMICAL ANALYSIS

Observable	Source	Role in the model
V_{obs}	MAPS	Two-dimensional line-of-sight velocity field
IVAR	MAPS	Measurement uncertainty, with $\sigma_{\text{meas}} = 1/\sqrt{\text{IVAR}}$
SNR	MAPS	Spaxel-quality screening
r, ϕ	MAPS	Elliptical polar-coordinate geometry of the velocity field; r is converted to physical kpc in the analysis pipeline
PA, b/a	MAPS	Position angle and photometric axis ratio used to define the empirical projection geometry
σ_{gas}	MAPS	Gas velocity-dispersion quality screening and reference for asymmetric-drift modeling
R_e	DRPALL	Prior scale for the disk and bulge structure
M_*	DRPALL	Global stellar-mass prior anchor in the mass decomposition
n	DRPALL	Morphology proxy and bulge-fraction prior
z	DRPALL	Cosmological scaling through $H(z)$ and the angular-diameter distance

Sources and roles follow the MaNGA data products described by SDSS-IV Collaboration (2024b,a) and K. B. Westfall et al. (2019).

empirical filter designed to assess whether the kinematics are organized enough for stable posterior inference—without attempting any mass decomposition. The projected velocity field is written as

$$V_{\text{obs}}(r, \phi) = V_{\text{sys}} + V_{\text{rot}}(r) \sin i \cos(\phi - \phi_0), \quad (1)$$

and the intrinsic empirical rotation curve is parameterized as

$$V_{\text{rot}}(r) = V_c \tanh\left(\frac{r}{R_t}\right) + s_{\text{out}} r. \quad (2)$$

Equation (1) is the standard thin-disk projection formula; because kinematic signal clusters near the major axis, only spaxels with $\text{SNR} \geq 10$ and azimuthal offsets $\leq 60^\circ$ from it are retained (M. Yang et al. 2021; E. O. Schmidt et al. 2023).

Equation (2) is a three-parameter tanh+linear form that captures the inner solid-body rise via (V_c, R_t) and a residual outer gradient s_{out} . Its role is diagnostic, not physical: a galaxy unable to constrain this simple model will not constrain the full baryon-plus-halo decomposition either.

After the spaxel-level screening, we fit the empirical model to each galaxy in a Bayesian framework. The free parameters are $V_c, R_t, s_{\text{out}}, V_{\text{sys}}$, and the geometric parameters i and $\Delta\phi$, which describe deviations around the photometric inclination and zero-point position angle. We also include an intrinsic-scatter term σ_{int} and the Student- t degrees-of-freedom parameter ν . Posterior sampling is then performed with the No-U-Turn Sampler (NUTS) in PyMC 5, and posterior-predictive diagnostics are used to assess how well the empirical model reproduces the velocity field. Specifically, the likelihood is written as

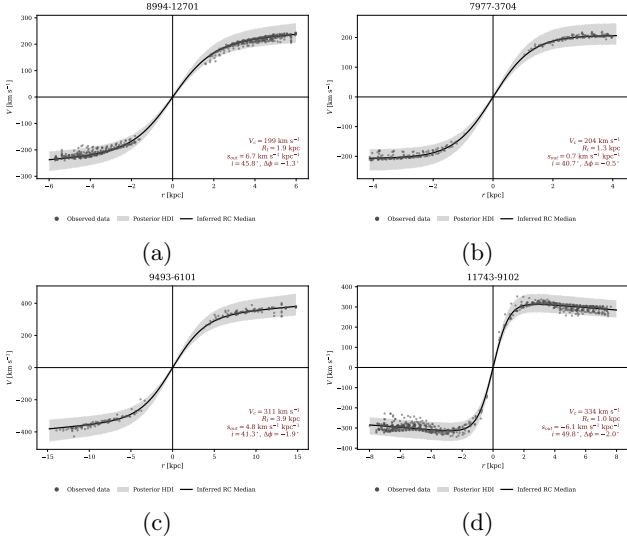


FIG. 1.— Bayesian empirical rotation-curve fits for four representative MaNGA galaxies. Points show H α rotation velocities after projection correction; solid curves and shaded regions show the posterior-median models and 95% HDIs of the rotation curves, respectively.

$$V_{\text{obs}} \sim \text{StudentT}(\nu, \mu = V_{\text{obs,model}}, \sigma = \sqrt{\sigma_{\text{meas}}^2 + \sigma_{\text{int}}^2}), \quad (3)$$

where $\sigma_{\text{meas}} = 1/\sqrt{\text{IVAR}}$. The Student- t form down-weights kinematically anomalous spaxels, keeping the screen sensitive to the global rotation pattern rather than individual outliers. Acceptance rests on posterior-predictive HDI coverage and overlap (thresholds in Section 2.3)—not on a single aggregate residual. Figure 1 shows representative outcomes for four galaxies spanning the sample’s range of Sérsic index and baryonic mass.

2.3. Kinematic Selection and Sample Definition

Galaxy-by-galaxy acceptance combines the empirical posteriors with the kinematic quality thresholds summarized as

$$\left\{ \begin{array}{l} 25^\circ \leq i \leq 70^\circ, \\ N_{\text{valid}} \geq 150, \\ R_{\text{out}}/R_t \geq 2, \\ \hat{R} \leq 1.05, \\ \text{ESS} \geq 200, \\ f_{\text{HDI}} > 0.60, \\ g_{\text{HDI}} > 0.80, \end{array} \right. , \quad (4)$$

where i is the photometric inclination, N_{valid} is the number of usable velocity spaxels, R_{out}/R_t measures the observational radial coverage, $\hat{R}$ is the Gelman–Rubin convergence statistic, ESS is the effective sample size, f_{HDI} is the fraction of observed velocities falling within the 95% posterior-predictive HDI, and g_{HDI} is the fraction of spaxels for which the measurement-uncertainty interval overlaps that same HDI. Of the 8070 galaxies in the parent inclination-cut sample, 1234 clear all seven thresholds and advance to the single-galaxy dynamical

inference of Section 3. Most of the 6836 rejected galaxies are removed because the empirical tanh+linear model fails to return a usable fit, indicating velocity fields too disordered for stable screening inference. Insufficient radial coverage ($R_{\text{out}}/R_t < 2$) is the second most common cause, accounting for about 17% of the rejected sample and reflecting the limited radial reach of the MaNGA bundles relative to the turnover radius.

3. SINGLE-GALAXY DYNAMICAL MODEL

The 1234 galaxies passing the empirical screen now enter full physical mass decomposition. Retaining the geometric projection framework of Section 2, we perform Bayesian inference over M_{200} , c , and the associated baryonic and nuisance parameters for each galaxy—writing out explicitly the stellar, NFW-halo, pressure-support, and error-model terms—and preserve the uncompressed posterior draws for the population analysis that follows.

The intrinsic rotational support is decomposed into baryonic, dark-matter, and asymmetric-drift terms according to

$$V_{\text{rot}}^2(r) = V_{\text{baryon}}^2(r) + V_{\text{dm}}^2(r) - V_{\text{drift}}^2(r). \quad (5)$$

In the current implementation, we approximate the baryonic term by the stellar contribution, that is, $V_{\text{baryon}} \approx V_*$. The gravitational contribution of gas to the rotation curve is not modeled as a separate term; the physical justification for this approximation in the context of the intermediate-mass disk galaxies studied here is discussed below. The kinematic effect of gas turbulence on ordered rotation does enter the model through the asymmetric-drift term listed in Table 2. Equation (5) is therefore not a complete hydrodynamical description, but a baseline force-balance prescription: the stellar and halo terms together provide circular support, while asymmetric drift subtracts the pressure-supported fraction from the ordered rotation. This parameterization is consistent with the code implementation and sufficient to separate the mass components best constrained by the current data, while propagating the resulting uncertainty structure to the population analysis.

3.1. Single-Galaxy Mass Model Components

We construct the single-galaxy model from three analytic components: a stellar mass term, an NFW halo term, and a pressure-support correction. Analytic forms are adopted because MaNGA velocity fields constrain only a limited radial range and must be modeled repeatedly at scale; compressing the force budget into a small number of physically interpretable parameters keeps sampling tractable while maintaining direct comparability with the literature, at the cost of representing a dynamical baseline rather than a complete mass reconstruction. Table 2 summarizes the adopted prescriptions.

The bulge+disk decomposition separates the centrally concentrated stellar contribution from the extended disk support, with $R_d = R_e/1.678$ and $a = R_e/1.8153$ tying both components to a single photometric scale; bars, thick disks, and M/L gradients are not modeled. We omit an explicit gaseous contribution to the gravitational potential: at $\log_{10}(M_*/M_\odot) \gtrsim 9.5$, combined HI+H₂ masses are typically $\lesssim 20\%$ of M_* (A. Saintonge & B. Catinella 2022), and gas disks are more radially extended

TABLE 2
ANALYTIC PRESCRIPTIONS FOR SINGLE-GALAXY MASS MODEL COMPONENTS

Component	Model	Comments
Rotation Curve	(5) $V_{\text{rot}}^2(r) = V_{\text{baryon}}^2(r) + V_{\text{dm}}^2(r) - V_{\text{drift}}^2(r)$	Baseline force-balance prescription for the single-galaxy mass decomposition; the stellar and halo terms provide circular support, while asymmetric drift subtracts pressure support from the ordered rotation.
Baryon		
Baryon component	(6) $V_{\star}^2(r) = V_{\text{bulge}}^2(r) + V_{\text{disk}}^2(r)$	The baryonic contribution is approximated by the stellar term alone, with $M_{\star}$ divided into bulge and disk components through f_{bulge} .
Hernquist bulge (L. Hernquist 1990)	(7) $V_{\text{bulge}}^2(r) = \frac{GM_{\text{bulge}}r}{(r+a)^2}$	$M_{\text{bulge}} = f_{\text{bulge}}M_{\star}$ and $a = R_e/1.8153$ in the adopted prior structure.
Exponential disk (K. C. Freeman 1970)	(8) $V_{\text{disk}}^2(r) = \frac{2GM_{\text{disk}}}{R_d}y^2 \times [I_0(y)K_0(y) - I_1(y)K_1(y)]$	$M_{\text{disk}} = (1 - f_{\text{bulge}})M_{\star}$, $y = r/(2R_d)$, and $R_d = R_e/1.678$.
Dark Halo		
NFW halo (J. F. Navarro et al. 1997)	(9) $V_{\text{dm}}^2(r) = V_{200}^2 \frac{\ln(1+cx) - \frac{cx}{1+cx}}{x \left[\ln(1+c) - \frac{c}{1+c} \right]}$ $x \equiv r/R_{200}$	M_{200} and c are the fitted halo parameters; R_{200} is defined under the overdensity-200 convention.
Virial scaling (J. F. Navarro et al. 1997)	(10) $V_{200} = (10GH(z)M_{200})^{1/3}$	Connects the fitted halo mass to the NFW circular-speed normalization.
Background cosmology (Planck Collaboration et al. 2020)	(11) $H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_{\Lambda}}$	The adopted cosmological values from Planck Collaboration et al. (2020) are $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_{\Lambda} = 0.685$.
Drift		
Asymmetric drift (J. Binney & S. Tremaine 2008)	(12) $V_{\text{drift}}^2(r) = 2\sigma_0^2 \left(\frac{r}{R_d} \right)$	σ_0 is the characteristic gas velocity-dispersion scale; the term provides a first-order pressure-support correction.

than the MaNGA coverage, suppressing their contribution to the inner rotation curve (F. Lelli et al. 2016; E. M. Di Teodoro et al. 2023). This approximation is most consequential for the most gas-rich, low-Sérsic objects, where unmodeled baryonic support may bias c upward (Section 5.4).

Each component is chosen for analytic tractability rather than morphological completeness. The Hernquist sphere (L. Hernquist 1990) yields a closed-form circular-speed expression with finite total mass—welcome for the inner bulge–halo decomposition—at the cost of spherical symmetry and the neglect of bars or nuclear substructure. Freeman’s thin exponential disk (K. C. Freeman 1970) likewise offers a two-parameter rotation curve that maps the photometric scale directly onto a stellar-support profile at intermediate radii; the price is a constant M/L ratio and the conflation of bar-induced non-circular motions with the mean circular speed.

The NFW rows (J. F. Navarro et al. 1997; Planck Collaboration et al. 2020) anchor the fits to standard cosmological language and supply the two parameters— M_{200} and c —that the population stage requires. Their limitation is radial reach. MaNGA typically resolves a few percent of the virial radius, well short of the transition where the NFW slope pivots from r^{-1} to r^{-3} ; consequently, M_{200} and c remain strongly degenerate in every single-galaxy posterior (Figure 3). The NFW profile is retained as a fiducial parameterization for literature comparability, not as a claim about inner halo universality.

The asymmetric-drift term (J. Binney & S. Tremaine 2008) deserves particular attention because it is the principal nuisance: without it, turbulent pressure in the gas disk masquerades as a mass deficit in the halo, systematically biasing c upward at fixed M_{200} . The linear ramp

$V_{\text{drift}}^2 = 2\sigma_0^2(r/R_d)$ follows from the thin-disk Jeans equation under the assumption that the gas-dispersion scale length equals $R_d/2$; a scale length of R_d would halve the correction, indicating a factor-of-two systematic whose amplitude depends on a quantity MaNGA does not directly constrain. Rather than calibrating σ_0 against the observed dispersion map—which would fold spaxel-level noise directly into the halo posterior—we fit it as a free parameter jointly with the mass model. The correction thereby absorbs the leading-order pressure signal without resolving its spatial structure: bar-induced peaks, spiral shocks, and H II complex emission are all collapsed into a single scale σ_0 , a simplification whose residual effects are taken up in Section 5.4.

Over the radial range MaNGA resolves, stellar normalization, bulge fraction, halo mass, and concentration are mutually compensating—raising $M_{\star}$ while lowering c traces nearly identical rotation curves. This is not a modeling defect but the observable content of the data; the posterior geometry it produces is exactly the structure that Section 4 inherits and exploits.

3.2. Bayesian Framework, Priors, and Likelihood

Projecting Equation (5) through the thin-disk geometry of Section 2 gives the model velocity field $V_{\text{obs,model}}(r, \phi | \Theta)$; the full parameter vector Θ — M_{200} , c , stellar-structure terms, systemic velocity, and nuisance scalings—is then inferred via posterior sampling. Point estimates and Gaussian surrogates are insufficient. The bulge–halo and disk–halo degeneracies in these velocity fields produce posterior ridges whose orientation is itself information: it encodes which axis in (M_{200}, c) space the data constrain most strongly, and that geometry is what propagates—intact—to Section 4.

The priors stabilize the single-galaxy fits without pre-imposing the population-level c - M relation. The most important constraint is the SHMR prior, whose anchoring relation is written as

$$\frac{M_\star}{M_{\text{halo}}} = 2N \left[\left(\frac{M_{\text{halo}}}{M_1} \right)^{-\beta} + \left(\frac{M_{\text{halo}}}{M_1} \right)^\gamma \right]^{-1}, \quad (13)$$

where M_1 , N , β , and γ follow B. P. Moster et al. (2013). The three primary priors on $M_\star$, M_{200} , and c (Table 3) regularize the dominant mass-scale degeneracy without injecting a population-level c - M trend into individual fits.

Inclination is pinned to the photometric estimate from Section 2 rather than sampled jointly—the degeneracy between inclination and the central logistic-weighting function is severe enough to destabilize even well-constrained targets; all remaining structural and nuisance priors anchor to NASA-Sloan Atlas (NSA) photometric quantities (Table 3; sensitivity tests in Appendix B).

The DAP inverse-variance maps encode only photon statistics; beam smearing, kinematic asymmetries, and model incompleteness all contribute mismatch that the formal noise budget omits. We therefore adopt the Student- t likelihood of Equation (3), in which σ_{int} absorbs excess scatter and heavy tails suppress the leverage of kinematically deviant spaxels. A radial logistic weight ($r_0 = 0.3 r_{\text{max}}$, transition width $0.1 r_{\text{max}}$, minimum central weight 0.3) additionally down-weights the innermost spaxels, where beam smearing most severely distorts the slope of the rotation curve and therefore the inferred concentration.

Posterior sampling uses PyMC 5 (O. Abril-Pla et al. 2023) with NUTS (nutpie backend)—500 tuning steps, 1000 draws, up to four chains, acceptance target 0.95; for a review of MCMC methodology in astrophysical contexts, see S. Sharma (2017). Raw posterior samples are retained throughout; medians and ETIs serve only as diagnostics, not as the input to Section 4. ArviZ (R. Kumar et al. 2019) handles posterior summaries and convergence statistics. Figure 2 shows four representative fits, each column tracing the inferential chain from optical morphology and velocity field through to the decomposed rotation curve and halo constraints.

3.3. Single-Galaxy Posterior Outputs for Population Inference

What reaches Section 4 is not a table of medians and uncertainties but the raw posterior draws themselves. The joint (M_{200}, c) posterior of a single galaxy typically occupies a narrow, curved ridge; in a handful of targets it shows mild bimodality. Reducing this to a covariance ellipse would erase exactly the curvature that distinguishes galaxies with similar medians but different constraints on the population slope—the variation that matters most for inferring α and σ_{int} . Figure 3 illustrates the range of posterior shapes across four representative galaxies: similar medians can coexist with markedly different ridge orientations and widths. Each retained MCMC draw is propagated as a sample point in (M_{200}, c) -space to the population likelihood of Section 4; full multi-parameter

covariance for one representative galaxy is documented in Appendix C.

We apply the following convergence and goodness-of-fit criteria to each single-galaxy posterior to determine whether it is suitable for the population model:

$$\left\{ \begin{array}{l} \hat{R} \leq 1.05, \\ \text{ESS}_{\text{bulk}} \geq 200, \\ \chi_\nu^2 \leq 2.0, \\ 0.1 \leq p_{\text{PPC}} \leq 0.9, \\ |\rho(c, M_{200})| \leq 0.85, \end{array} \right. \quad (14)$$

where $\hat{R}$ is the Gelman–Rubin convergence statistic evaluated across all sampled parameters, ESS_{bulk} is the bulk effective sample size, χ_ν^2 is the reduced chi-squared of the posterior-median rotation-curve model, p_{PPC} is the posterior predictive check p-value, and $\rho(c, M_{200})$ is the Pearson correlation coefficient between the $\log c$ and $\log M_{200}$ posterior samples for a given galaxy. Of the 1234 galaxies entering the single-galaxy inference, 620 satisfy all criteria and are retained for the population-level analysis. The attrition from 1234 to 620 breaks down as follows: 289 galaxies fail to produce usable posteriors because of MCMC non-convergence or processing failure, leaving 945 converged fits. Among those, $\chi_\nu^2 > 2.0$ is the most selective quality cut and by itself removes about two-thirds of the 325 additional rejections, indicating model mismatch; most of the remainder are excluded by $|\rho(c, M_{200})| > 0.85$, consistent with posterior degeneracy under limited radial coverage. We show the distributions of redshift, stellar mass, inclination, and Sérsic index at each stage in Figure 7.

Table 4. Galaxy Counts at Each Stage of the Analysis Pipeline

Stage	Selection criteria	Number
Full sample	Galaxies satisfying the inclination cut $25^\circ \leq i \leq 70^\circ$	8070
Empirically screened sample	Galaxies passing Equation (4) and entering the physical dynamical model	1234
Final population-fit sample	Galaxies satisfying Equation (14) after the single-galaxy dynamical inference	620

4. POPULATION INFERENCE OF THE c - M RELATION

4.1. c - M Relation Parameterization

Six hundred and twenty per-galaxy posterior chains—each a probability distribution over (M_{200}, c) —are now the input to the population stage. Rather than fitting a line through point estimates, we ask what hyperparameter set Φ is consistent with the entire collection of single-galaxy posterior geometries taken together. Over the limited mass and redshift baseline of the sample, the log-linear parameterization of J. F. Navarro et al. (2009); A. A. Dutton & A. V. Macciò (2014) is adequate; defining the pivot $M_{\text{pivot}} = M_{\text{pivot}, h^{-1}}/h$ (with $M_{\text{pivot}, h^{-1}} = 10^{12} h^{-1} M_\odot$), we write the relation as

$$\log_{10} c = \log_{10} c_0 + \alpha \log_{10} \left(\frac{M_{200}}{M_{\text{pivot}}} \right). \quad (15)$$

Within this framework, we assign each galaxy i a latent-variable vector $\theta_i \equiv \{m_i, \ell_i\}$, where $m_i \equiv$

TABLE 3
PRIOR DISTRIBUTIONS IN THE SINGLE-GALAXY BAYESIAN MASS MODEL

Parameter	Distribution (width σ)	Role
$\log_{10} M_{\star}$	$\mathcal{N}(\log_{10} M_{\star, \text{NSA}}, 0.05 \text{ dex})$	NSA photometric stellar mass
$\log_{10} M_{200}$	$\mathcal{TN}(\mu_{\text{SHMR}}, 0.15 \text{ dex}; \pm 3\sigma)$	SHMR anchor; stabilizes $M_{200}-c$ degeneracy
$\log_{10} c$	$\mathcal{N}(\log_{10} 9.0, 0.50 \text{ dex})$	Independent; no $c-M$ slope imposed
$\log_{10} \sigma_0$	$\text{LogN}(\log(10 \text{ km s}^{-1}), 0.20 \text{ dex})$	Gas pressure-support scale
V_{sys}	$\mathcal{TN}(V_{\text{sys, NSA}}, 5 \text{ km s}^{-1}; \pm 20 \text{ km s}^{-1})$	Systemic velocity (geometric anchor)
i	$\delta(i - i_{\text{NSA}})$	Inclination fixed to photometric NSA value
$\log_{10} R_e$	$\text{LogN}(R_{e, \text{NSA}}, 0.05 \text{ dex})$	Photometric half-light radius
$\text{logit}(f_{\text{bulge}})$	$\mathcal{N}(1.2(n - 2.5), 0.2)$	Sérsic- n morphology prior
σ_{int}	$\text{Exp}(2\bar{\sigma}_{\text{meas}})$	Residual scatter beyond noise
$\nu - 2$	$\Gamma(2.0, 0.1)$	Student- t tail weight

NOTE. — $\mathcal{N}$: Normal; $\mathcal{TN}$: Truncated Normal (clipped at $\pm 3\sigma$ unless noted); LogN : LogNormal; Exp : Exponential. μ_{SHMR} is derived from $M_{\star, \text{NSA}}$ via B. P. Moster et al. (2013). The $\text{logit}(f_{\text{bulge}})$ prior is centered on the Sérsic- n logit-linear relation. See Appendix B for the inclination prior and M_{200} prior width sensitivity test.

$\log_{10} M_{200, i} - \log_{10} M_{\text{pivot}}$ and $\ell_i \equiv \log_{10} c_i$. The population distribution $p_{\text{pop}}(\theta_i | \Phi)$ is then specified as a factored Student- t model: the halo masses follow a Student- t marginal and the concentrations at fixed mass follow a Student- t conditional about the mean relation, both sharing the same degrees of freedom ν ,

$$m_i \sim t_{\nu}(\mu_M, \sigma_M^2), \quad \ell_i | m_i \sim t_{\nu}(\log_{10} c_0 + \alpha m_i, \sigma_{\text{int}}^2). \quad (16)$$

The hyperparameter set $\Phi = \{\mu_M, \sigma_M, c_0, \alpha, \sigma_{\text{int}}, \nu\}$ encodes both the mass distribution of the sample and the $c-M$ trend: c_0 fixes the pivot-mass normalization, α the slope, σ_{int} the intrinsic width, and ν the tail heaviness—a degree of freedom that allows the model to accommodate outliers without distorting the bulk inference. As $\nu \rightarrow \infty$ the Student- t marginals collapse to Gaussians; finite ν is anticipated by the PSIS diagnostics of Section 5.4, which show that a substantial fraction of the sample carries heavy importance-weight tails. Priors on Φ are weakly informative—anchored loosely to A. A. Dutton & A. V. Macciò (2014) but wide enough for the data to dominate—and are listed in full in Table 5.

A selection caveat attaches to every number in what follows. The 620 galaxies reaching this stage survived two successive quality cuts—the empirical kinematic screen of Section 2.2 and the posterior-quality thresholds of Section 3.3—so the inferred hyperparameters describe halos hosting *kinematically well-ordered* disk galaxies, not the halo population at large. Quoting c_0 , α , and σ_{int} without that qualification would misrepresent the scope of the measurement.

4.2. Prior-Corrected Population Likelihood from Posterior Samples

Strongly non-Gaussian (M_{200}, c) posteriors cannot be faithfully collapsed into covariance matrices without biasing the inferred population relation (D. W. Hogg et al. 2010); a sample-based approach is required. We propagate the stored posterior draws through a prior-corrected importance-sampling likelihood (S. Sharma 2017).

We obtain the exact marginal likelihood for galaxy i by integrating over its latent physical parameters θ_i :

$$\mathcal{L}_i(\Phi) = \int p(D_i | \theta_i) p_{\text{pop}}(\theta_i | \Phi) d\theta_i, \quad (17)$$

TABLE 5
PRIOR DISTRIBUTIONS FOR POPULATION-LEVEL $c-M$ HYPERPARAMETERS

Parameter	Prior specification	Comments
μ_M	$\mathcal{N}(\bar{m}_{\text{obs}}, 1.0^2)$	$\bar{m}_{\text{obs}}$ = sample mean of $\log_{10} M_{200} - \log_{10} M_{\text{pivot}}$; broad mass-center prior
σ_M	$\mathcal{HN}(1.0)$	Positive half-normal; broad prior on halo-mass spread of the sample
$\log_{10} c_0$	$\mathcal{N}(0.905, 0.5^2)$	Anchored to the A. A. Dutton & A. V. Macciò (2014) normalization; intentionally broad prior on the relation normalization
α	$\mathcal{N}(-0.101, 0.3^2)$	Centered on the A. A. Dutton & A. V. Macciò (2014) slope; $\sigma = 0.3$ is wide enough for the data to update
$\ln \sigma_{\text{int}}$	$\mathcal{N}(\ln 0.15, 0.8^2)$	Broad log-prior on the intrinsic scatter of the population relation
ν	$\text{Gamma}(2.0, 0.1)$	Prior mean ≈ 20 , allowing heavy-tailed ($\nu \approx 2-5$) to near-Normal behaviour

where D_i denotes the MaNGA kinematic data for that galaxy. Re-running the single-galaxy likelihood for each of 620 galaxies inside the population sampler is computationally out of reach. We therefore reuse the single-galaxy posterior as the importance-sampling proposal: since $p(\theta_i | D_i) \propto p(D_i | \theta_i) p_{\text{stage1}}(\theta_i)$, the marginal likelihood reduces to an expectation over that posterior, discretized over the S_i retained MCMC samples $\{\theta_{is}\}_{s=1}^{S_i}$.

Defining the raw log-importance-weights $\log w_{is} = \log p_{\text{pop}}(\theta_{is} | \Phi) - \log p_{\text{stage1}}(\theta_{is})$, where p_{stage1} is the prior used in the single-galaxy inference, each term reweights a posterior draw by the population-to-prior density ratio, removing the stage-1 prior imprint under the candidate $c-M$ model. When the single-galaxy posterior and the population model differ substantially, a few samples can receive extreme weights, inflating the variance of the estimate. We therefore apply per-galaxy truncated importance sampling, capping each weight at a per-galaxy threshold τ_i that bounds the effective weight

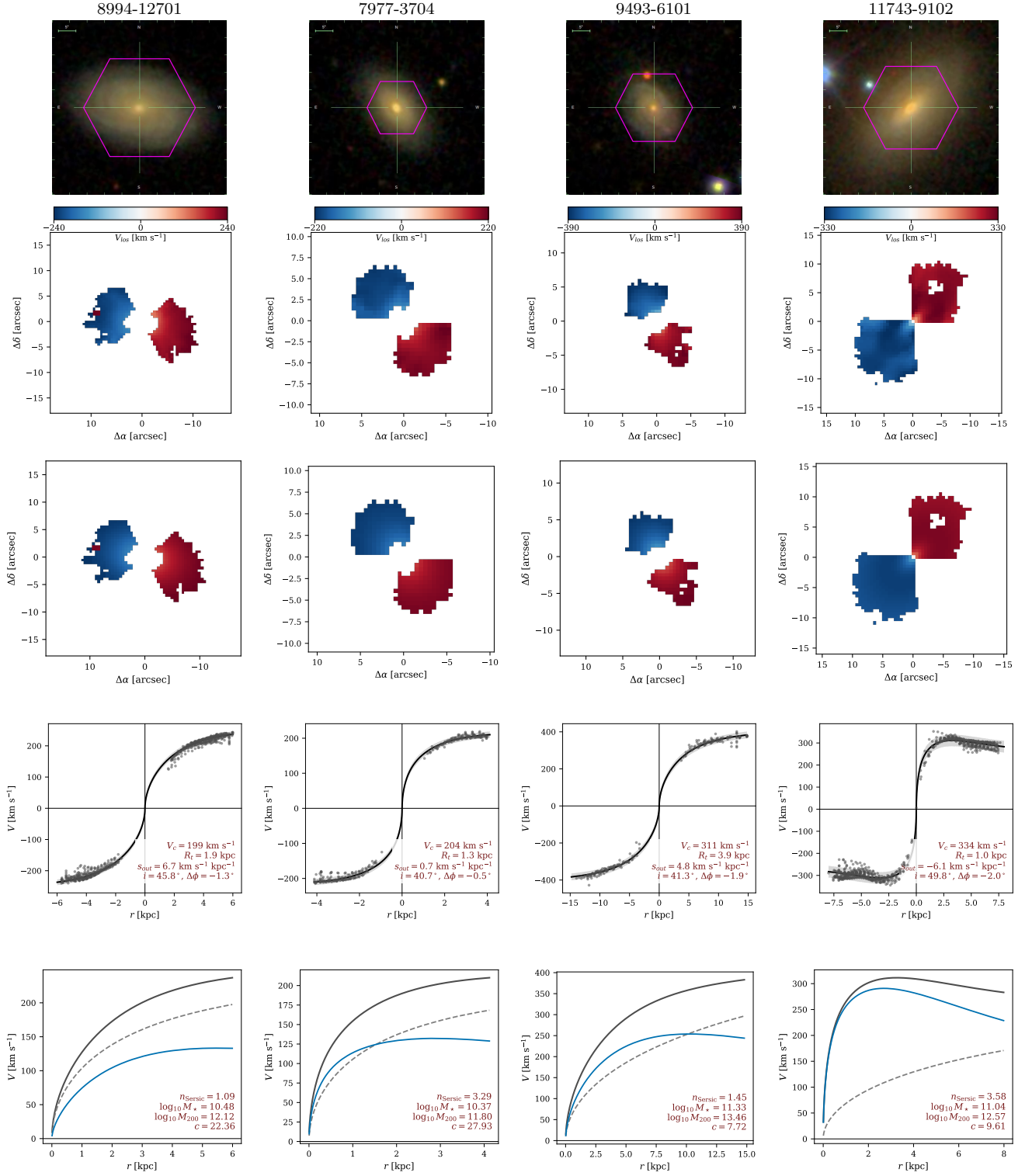


FIG. 2.— Single-galaxy NFW dynamical fits for four representative MaNGA galaxies. Each column corresponds to one galaxy; rows show the optical image, observed velocity field, modeled velocity field, rotation curve with 95% ETI, and decomposition into total, stellar, and dark-matter components. Together the panels illustrate how two-dimensional kinematic constraints map into the halo-parameter posteriors propagated to the population model.

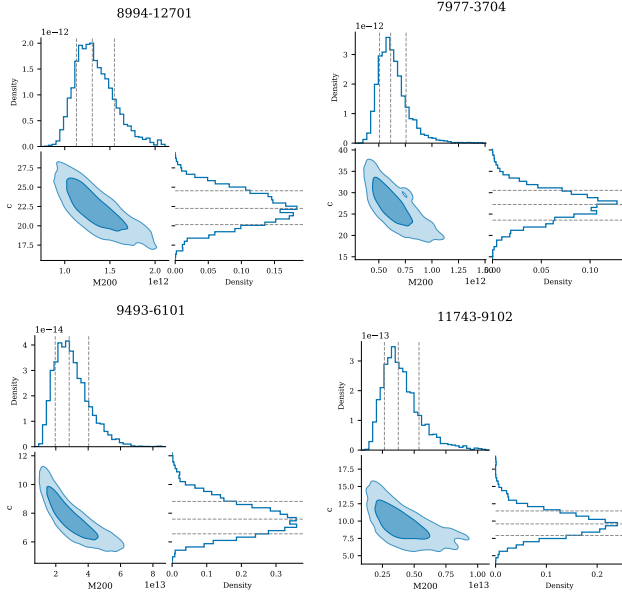


FIG. 3.— Joint M_{200} - c posterior distributions for four representative galaxies spanning low/high Sérsic n and low/high M_* combinations. Single-galaxy posteriors trace narrow, curved degeneracy ridges in the M_{200} - c plane, motivating direct propagation of uncompressed posterior samples to the population model. variance while introducing only $\mathcal{O}(S_i^{-1/2})$ bias, giving the truncated estimator

$$\log \mathcal{L}_i(\Phi) \approx \text{logsumexp}_s [\min(\log w_{is}, \log \tau_i)] - \log S_i. \quad (18)$$

The galaxy-level priors in Section 3.2 are designed to provide only mild regularization, and the pipeline retains the uncompressed posterior draws so that the true measurement geometry can be propagated as faithfully as possible.

At the population level, we infer the hyperparameters by joint posterior sampling so that the correlated uncertainties among the normalization, slope, and intrinsic scatter are preserved. Figure 4 shows the prior-to-posterior updating for the population normalization and slope.

5. RESULTS AND DISCUSSION

5.1. Inferred c - M Relation for the Full Sample

We first fit the baseline population model to the full retained sample of 620 galaxies using the prior-corrected posterior-sample likelihood described in Section 4.2, with 1000 posterior samples drawn per galaxy. Table 6 lists the posterior parameter values. Over the mass range covered by the current data, the sample supports a c - M relation with a modest negative slope when the photometric inclinations are held fixed. As noted in Section 4.1, these results are conditional on the kinematic selection function: they describe the c - M relation of halos hosting late-type disk galaxies that satisfy our screening criteria, not an unconditional constraint on the full halo population.

We summarize this result more directly in Figure 5. The gray cloud represents randomly selected draws from the single-galaxy (M_{200}, c) posteriors; for each of the 620 retained galaxies we plot a fixed-size random subset of posterior samples, so the cloud directly visualizes

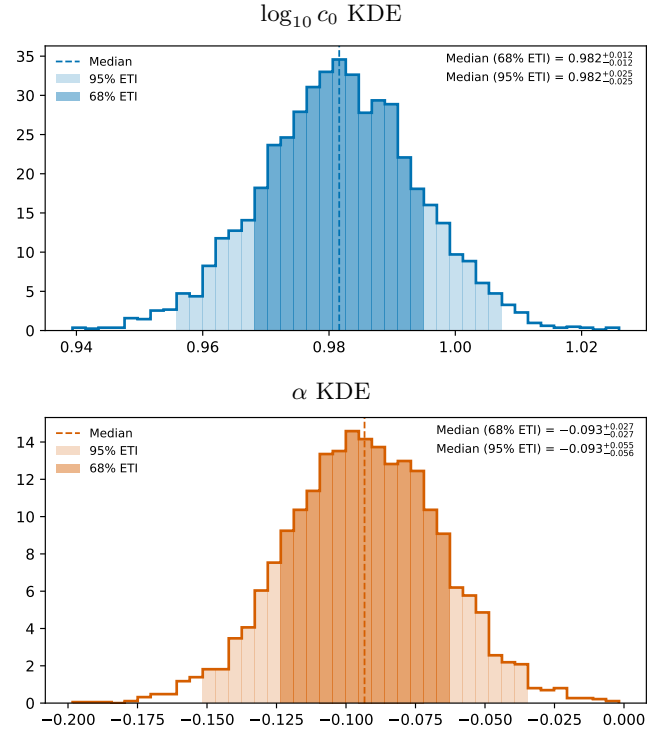


FIG. 4.— Prior-to-posterior updating for population c - M hyperparameters. *Upper*: one-dimensional kernel density estimate (KDE) and prior distribution for the normalization $\log_{10} c_0$. *Lower*: corresponding diagnostics for the slope α . Together the panels illustrate how the MaNGA kinematic data update each hyperparameter relative to its prior.

TABLE 6
INFERRED AND LITERATURE c - M RELATION PARAMETERS

Source	$\log_{10} c_0$	α	Scatter (dex)
This work ^a	$0.982^{+0.023}_{-0.024}$	$-0.093^{+0.051}_{-0.053}$	$0.237^{+0.027}_{-0.026}$
Dutton & Macciò 2014 ^b	0.905	-0.101	0.11
Li et al. 2020 ^c	0.840 ± 0.030	-0.060 ± 0.040	0.20
Yasin et al. 2023 ^d	0.910 ± 0.050	-0.110 ± 0.030	0.15

^a 95% ETI; superscripts and subscripts give the upper and lower offsets from the median.

^b A. A. Dutton & A. V. Macciò (2014); no uncertainty is quoted in the source paper.

^c P. Li et al. (2020); $\pm$ values are 1σ .

^d T. Yasin et al. (2023); $\pm$ values are 1σ .

the uncompressed posterior geometry that the population model receives as input. The solid blue curve shows the posterior-median relation from our baseline population model, indicating that concentration decreases gradually with increasing halo mass over the sampled mass interval. For comparison, the reference relation of A. A. Dutton & A. V. Macciò (2014) predicts a steeper negative slope, but the two remain consistent in the sign of the trend. The figure also shows the $\pm 2\sigma$ band of the reference relation using $\sigma_{\log c} = 0.11$ dex in order to illustrate the typical scatter allowed by the numerical-simulation calibration itself. The blue shaded band around the fitted relation represents the full allowed population region obtained by combining the 95% ETI uncertainties in c_0 and α with σ_{int} in logarithmic space.

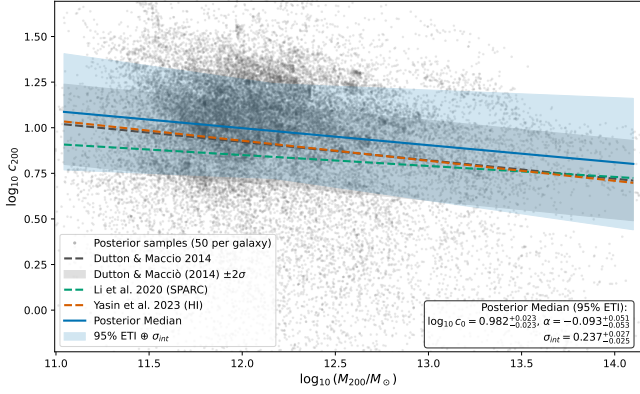


FIG. 5.— Halo c - M relation for the full sample of 620 MaNGA disk galaxies. Gray points are randomly selected draws from the stage-1 (M_{200} , c) posterior of each galaxy; they visualize the posterior support of individual halos and are not independent galaxy measurements. The solid blue curve and shaded blue band show the posterior-median relation and its 95% ETI $\oplus \sigma_{\text{int}}$ region (see Table 6). Literature comparisons are shown as a dark-gray dashed curve with a dark-gray $\pm 2\sigma$ band for A. A. Dutton & A. V. Macciò (2014), a green dashed curve for P. Li et al. (2020), and an orange dashed curve for T. Yasin et al. (2023); parameter values are listed in Table 6.

5.2. Comparison with Simulations and Published Results

The headline result is $\alpha = -0.093$, with a 95% ETI of $[-0.146, -0.042]$ that cleanly excludes zero. It reproduces the anti-correlation that N-body simulations have consistently predicted: later-forming halos collapse at lower background density, depressing concentrations for progressively more massive systems (R. H. Wechsler et al. 2002; J. F. Navarro et al. 2009; T. Ishiyama et al. 2021). The sign is correct; the exclusion of zero is not marginal.

The normalization $\log_{10} c_0 = 0.982$ sits 0.077 dex above A. A. Dutton & A. V. Macciò’s 2014 reference value of 0.905; the slope $\alpha = -0.093$ is shallower than their -0.101 . Neither discrepancy exceeds the width of the respective uncertainty bands, and the dark-gray $\pm 2\sigma$ envelope around A. A. Dutton & A. V. Macciò’s relation ($\sigma_{\log c} = 0.11$ dex, fixed) and the blue ETI-plus- σ_{int} band from the present fit overlap substantially across the sampled mass interval. The crucial distinction is not in the overlap but in the provenance of the two bands: one encodes a scatter assumption extracted from a dark-matter-only halo catalog; the other is inferred from 620 kinematically filtered IFU posteriors, each carrying its own non-Gaussian covariance structure.

What the agreement demonstrates is methodological robustness rather than parameter concordance. A negative slope survives even when non-Gaussian single-galaxy posteriors replace point estimates and an explicit σ_{int} replaces the tacit assumption of negligible scatter—a less trivial statement than it may appear. The same data, compressed into medians and covariance matrices before the population fit, would not guarantee the same sign.

5.3. Physical Interpretation of the Recovered Relation

The sign of α encodes the late-assembly trend (R. H. Wechsler et al. 2002; J. F. Navarro et al. 2009). Its magnitude—0.093, shallower than A. A. Dutton & A. V. Macciò’s simulation-calibrated 0.101 by roughly 8%—is more ambiguous; the data alone do not determine its origin, and neither mechanism discussed below is directly

modeled or tested against hydrodynamical mock observations in this work.

Selection is the primary suspect. Disk-dominated, late-type systems—the only objects this pipeline retains (Appendix A)—inhabit halos assembled predominantly through sustained cold-gas accretion; their formation-epoch distribution is narrower than the full relaxed-halo population underlying the A. A. Dutton & A. V. Macciò (2014) calibration. A narrower formation-epoch span maps to a compressed concentration range, directly attenuating the inferred slope.

Baryonic physics offers a second, mass-dependent set of biases that act in opposing directions. Supernova-driven outflows suppress the NFW-equivalent c at $M_{200} \lesssim 10^{11.5} M_{\odot}$ by converting the inner cusp into a core (A. Pontzen & F. Governato 2012); dissipative disk settling and adiabatic contraction raise it at the high-mass end (G. R. Blumenthal et al. 1986; P. Li et al. 2022). Their net imprint on α is a prediction that only a hydrodynamical mock-recovery test could quantify.

The broad allowed region in Figure 5 shows that the recovered c - M relation carries $\sigma_{\text{int}} = 0.237$ dex intrinsic scatter, roughly twice the 0.11 dex found in dark-matter-only simulations (A. A. Dutton & A. V. Macciò 2014). MaNGA’s limited radial coverage leaves the single-galaxy (M_{200} , c) posterior unresolved; σ_{int} therefore absorbs both genuine halo-to-halo cosmological scatter and unresolved single-galaxy measurement uncertainty, rather than representing the intrinsic cosmological scatter alone. The quoted $\sigma_{\text{int}} = 0.237$ dex is therefore best read as an upper bound on the true cosmological scatter for this kinematically selected population—it cannot be compared directly to the 0.11 dex isolation of purely cosmological variation reported by A. A. Dutton & A. V. Macciò (2014). A secondary contribution may arise from genuine diversity in baryonic assembly histories—differences in feedback efficiency, gas fraction, and merger history across the sample—that the present framework does not resolve. Disentangling the two contributions would require either a mock-recovery test with known input scatter propagated through the full pipeline, or a fully joint hierarchical model in which the single-galaxy inference is embedded within the population stage; both are deferred to future work.

Since MaNGA velocity fields constrain the total gravitational response of bulge, disk, halo, and pressure support jointly, the M_{200} - c degeneracy in the single-galaxy posteriors is itself observational information rather than a proxy for the pure dark-matter relation. The 620 galaxies entering the population fit are, by construction, kinematically regular, disk-dominated objects with lower Sérsic indices and intermediate inclinations—systematically different from the relaxed-halo population underlying the A. A. Dutton & A. V. Macciò (2014) calibration regardless of morphology. The recovered c - M relation therefore applies to halos hosting nearby late-type galaxies with well-ordered rotational support, not to the full Λ CDM halo population.

5.4. Systematic Effects and Limitations

The present framework is a controlled baseline rather than an exhaustive dynamical description. Beam smearing is mitigated by logistic down-weighting within $0.3 r_{\text{max}}$ rather than explicit point spread function (PSF)

convolution; because concentration is encoded in the steepness of the inner rotation curve, this approximation may bias c downward for genuinely high-concentration halos. Conversely, omitting an explicit gas-mass term can bias c upward at the low-mass, gas-rich end of the sample, as unmodeled baryonic support is absorbed by the NFW halo (Section 3.1); a 10–20% uncertainty in σ_0 shifts V_{rot}^2 by $\lesssim 1\text{--}3\%$, subdominant for most galaxies. These biases act in opposite directions; their net effect on α requires a full perturbation test, deferred to future work.

The normalization c_0 inherits sensitivity to the adopted B. P. Moster et al. (2013) SHMR prior, which is necessary to anchor the halo-mass scale under the limited radial coverage of MaNGA, and could shift if a different calibration (e.g., P. S. Behroozi et al. 2013) or wider prior width were used. The slope α is less susceptible, as it is driven by the differential variation of concentration across the mass range with the concentration prior set independently; formal sensitivity tests are provided in Appendix B.

The framework also assumes the NFW profile with its fixed inner cusp $\rho \propto r^{-1}$. Supernova-driven outflows can convert this cusp into a core in intermediate-mass halos (A. Pontzen & F. Governato 2012); an NFW fit then compensates by reducing c , and because core-formation efficiency is mass-dependent, this could bias both c_0 and α . At the high-mass end, dissipative bulge settling tends instead to contract the halo (G. R. Blumenthal et al. 1986; P. Li et al. 2022), biasing c in the opposite direction. The recovered c – M relation should therefore be understood as a parameterization of the effective NFW-equivalent halo structure of the sample.

The Pareto Smoothed Importance Sampling (PSIS) diagnostic (A. Vehtari et al. 2024) quantifies the tail behavior of the importance weights $w_{is} = p_{\text{pop}}(\theta_{is} | \Phi) / p_{\text{stage1}}(\theta_{is})$ used in the population reweighting (Section 4.2). Of the 620 galaxies, 262 (42.3%) return $\hat{k} < 0.5$ —the reliable regime—while 122 (19.7%) fall in the caution band $0.5 \leq \hat{k} < 0.7$ and 236 (38.1%) exceed the unreliability threshold $\hat{k} \geq 0.7$. Figure 6 highlights why this demands attention: for 57.7% of the sample the importance-weight variance is formally infinite, and for the 236 objects at $\hat{k} \geq 0.7$ the reweighting is plainly unreliable, capable of locally over- or under-weighting individual galaxies in ways the truncation heuristic can only partially contain. The root cause is MaNGA’s limited radial reach—broad single-galaxy (M_{200}, c) posteriors whose ridge orientation diverges from the narrow population locus generate heavy reweighting tails almost automatically. These galaxies are downweighted rather than discarded, so the aggregate inference is not formally invalidated; but the fraction is large enough that an iterated importance-sampling scheme or a fully joint hierarchical sampler—in which the single-galaxy and population stages share a common posterior—would be the principled remedy, and is deferred to future work.

These systematics collectively bound the scope of the present results: the recovered c – M relation is a probabilistic constraint under the current observational and modeling assumptions, not a definitive measurement of the underlying halo population. The most natural exten-

sions are explicit PSF convolution, an explicit gas-mass term, and multi-model comparison at the single-galaxy stage. At the extreme high-concentration tail, a small number of outliers deviate from the median relation, as seen in Figure 9, and may signal profile departures that the present framework cannot resolve (D. Kong & H.-B. Yu 2025).

6. SUMMARY AND CONCLUSIONS

Applied to 620 kinematically screened MaNGA late-type disk galaxies, the three-stage Bayesian pipeline yields $\log_{10} c_0 = 0.982^{+0.023}_{-0.024}$, $\alpha = -0.093^{+0.051}_{-0.053}$, and intrinsic scatter $\sigma_{\text{int}} = 0.237^{+0.027}_{-0.026}$ dex (95% ETI). Under fixed photometric inclinations the ETI for α lies entirely negative—a Λ CDM-consistent anti-correlation for this kinematically selected population—but the conclusion is fragile: a $\sigma_i = 5^\circ$ Gaussian inclination prior widens the interval to straddle zero (Appendix B), so the significance of the recovered slope is conditional on the fidelity of the photometric axis-ratio estimates.

The central methodological lesson is unambiguous: posterior information must reach the population stage in sample form. MaNGA velocity fields yield (M_{200}, c) constraints whose geometry is curved, covariant, and heterogeneous across galaxies; compressing that geometry into point estimates or Gaussian approximations before the population fit discards exactly the information that distinguishes a slope from a normalization offset. The recovered c – M relation is a baseline for nearby late-type galaxies with well-ordered rotational support, not a general calibration of the halo c – M relation.

The contraction from 8070 galaxies to 620—a factor of thirteen—is itself a diagnostic: it marks the current boundary of where the pipeline is self-consistent. Within that boundary, the single-galaxy stage rests on an idealized bulge-plus-disk stellar model, an NFW halo, and a single-scale pressure-support correction, without explicit PSF convolution, a gas-mass term, or alternative halo profiles. Two extensions are the natural next steps: broadening the retention fraction through more flexible single-galaxy models, and embedding the single-galaxy and population stages in a jointly sampled hierarchy. Both are prerequisites for the present framework to graduate from a methodological demonstration to a physically interpretable observational calibration.

ACKNOWLEDGMENTS

We thank Professor [Name] (Princeton University) for stimulating discussions and for sharing insights that helped shape the physical motivation and conceptual framing of this work. We are grateful for continued guidance during the course of this project.

This work relied on open-source scientific software for Bayesian modeling, numerical computation, data handling, and plotting. We thank the developer communities behind these tools for making the analysis possible.

Software: PyMC 5 (O. Abril-Pla et al. 2023), ArviZ (R. Kumar et al. 2019), nutpie (A. Seyfried 2023), NumPy (C. R. Harris et al. 2020), SciPy (P. Virtanen et al. 2020), Astropy (Astropy Collaboration et al. 2022), Matplotlib (J. D. Hunter 2007), pandas (W. McKinney 2010)

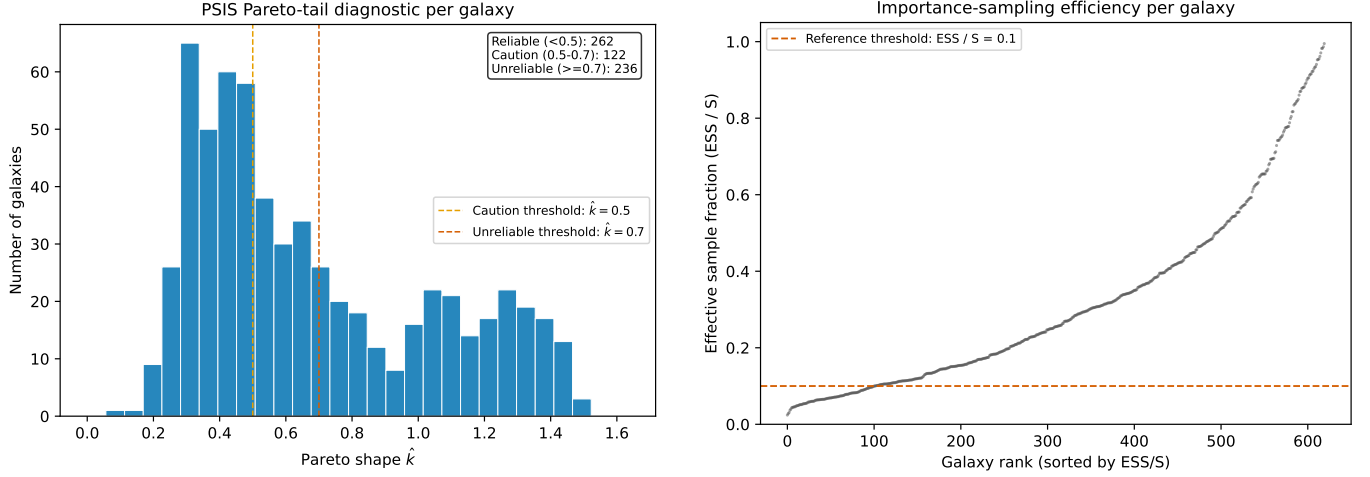


FIG. 6.— PSIS diagnostics for the 620 retained galaxies at the posterior-median hyperparameters Φ^* . *Left*: distribution of the Pareto shape parameter $\hat{k}$; dashed lines mark the caution ($\hat{k} = 0.5$, orange) and unreliability ($\hat{k} = 0.7$, red-orange) thresholds with per-regime counts annotated. *Right*: ESS/S per galaxy sorted in ascending order; the dashed orange line marks the reference threshold ESS/S = 0.1.

CODE AVAILABILITY

Source code and reproducible notebooks are available on GitHub at <https://github.com/Planetarian39/manga-dm>.

APPENDIX

GALAXY SAMPLE PROPERTIES ACROSS SELECTION STAGES

This appendix documents how the final sample differs from the broader parent samples in its basic observed properties. Figure 7 compares the redshift, stellar-mass, inclination, and Sérsic-index distributions for the full sample, the empirically screened sample, and the final population-level sample of 620 galaxies. The redshift distribution changes little across the three stages, and the stellar-mass distribution shifts only slightly, confirming that the screening does not strongly favor any particular distance range or mass scale. The inclination and Sérsic-index distributions change more sharply: the final sample concentrates at intermediate inclinations and at lower Sérsic n , as expected for rotationally dominated systems. The sharpest contrast appears at Sérsic $n > 5$: the parent sample contains many such objects, whereas the final retained sample contains very few, because the bulge-plus-disk model fits poorly to pressure-supported, bulge-dominated galaxies. High-Sérsic- n objects would require either a more general pressure-support parametrization or a mass-decomposition scheme designed for early-type morphologies. The corresponding sample counts are summarized in Table 4, while the detailed definitions of the screening conditions, empirical model, and retention thresholds are given in Section 2 of the main text.

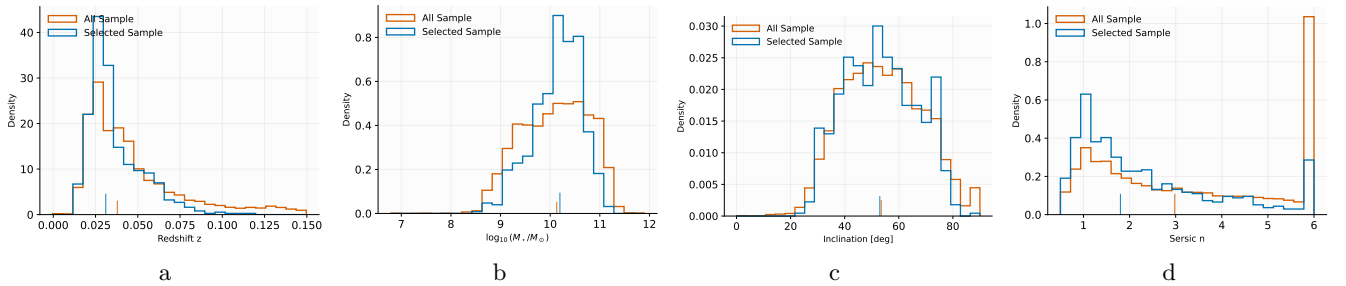


FIG. 7.— Galaxy sample properties at each stage of the analysis pipeline. The panels show, from left to right, the redshift, stellar mass, inclination, and Sérsic-index distributions for the full sample, the empirically screened sample, and the final population-fit sample. The final retained sample changes only modestly in redshift and stellar mass, but becomes more strongly weighted toward dynamically regular, more clearly rotation-supported, disk-dominated systems in inclination and Sérsic index.

SENSITIVITY TO PRIOR CHOICES AND MODEL ASSUMPTIONS

We test the robustness of the population hyperparameters ($\log_{10} c_0$, α , σ_{int}) against three modeling choices on a representative $N = 60$ subsample. The baseline uses fixed inclination ($i = i_{\text{NSA}}$), $\sigma_{\log M_{200}} = 0.15$ dex, and inner logistic down-weighting at $r_0 = 0.3 r_{\text{max}}$. Test (i) adds a Gaussian inclination prior ($\sigma_i = 5^\circ$); test (ii) doubles the SHMR prior width to 0.30 dex; test (iii) extends the beam-smearing exclusion to $r_0 = 0.5 r_{\text{max}}$. Results are in Table 7.

TABLE 7
SENSITIVITY OF c - M HYPERPARAMETERS TO MODEL CHOICES (95% ETI; $N = 60$ SUBSAMPLE FROM 620 GALAXIES).

Test	σ_i	$\sigma_{\log M_{200}}$	$r_0/r_{\max}$	$\log_{10} c_0$	α	σ_{int} (dex)
Baseline	fixed	0.15 dex	0.3	$1.026^{+0.069}_{-0.079}$	$-0.134^{+0.137}_{-0.140}$	$0.225^{+0.099}_{-0.067}$
(i) Inc prior	5°	0.15 dex	0.3	$0.875^{+0.175}_{-0.199}$	$-0.174^{+0.287}_{-0.306}$	$0.373^{+0.192}_{-0.135}$
(ii) M_{200} prior	fixed	0.30 dex	0.3	$1.153^{+0.186}_{-0.235}$	$-0.147^{+0.333}_{-0.288}$	$0.145^{+0.197}_{-0.080}$
(iii) Beam smearing	fixed	0.15 dex	0.5	$1.029^{+0.070}_{-0.079}$	$-0.140^{+0.140}_{-0.138}$	$0.225^{+0.108}_{-0.071}$

Test (i) is the dominant systematic: it shifts $\log_{10} c_0$ by -0.151 dex and inflates σ_{int} from 0.225 to 0.373 dex, while the ETI for α widens to encompass zero. Test (ii) shifts $\log_{10} c_0$ upward by $+0.127$ dex and reduces σ_{int} to 0.145 dex, but leaves α nearly unchanged ($\Delta\alpha = -0.013$), confirming that the negative slope is insensitive to the SHMR prior width. Test (iii) produces negligible shifts ($\Delta\log_{10} c_0 = +0.003$ dex, $\Delta\alpha = -0.006$), so the inner down-weighting radius has no meaningful effect.

Fixing the inclination to the photometric value is an operational baseline, not a statistically conservative choice: it isolates the effects of the other two model parameters on the inferred slope, but it suppresses a physically motivated source of uncertainty. The credible intervals in Table 6 are therefore conditional on the accuracy of the photometric inclinations and should be regarded as lower bounds on the true parameter uncertainties. Improving inclination characterization—through higher-resolution photometric axis-ratio measurements or joint kinematic–photometric decomposition—is the highest-priority methodological extension of the present baseline.

POSTERIOR DIAGNOSTICS FOR A REPRESENTATIVE GALAXY

This appendix illustrates the quality of the single-galaxy inference using galaxy 11743-9102. The parameters shown include stellar mass, halo parameters, systemic velocity, bulge fraction, pressure-support parameter, and structural scale.

Figure 8 shows the posterior diagnostics. Panel (a) shows the MCMC trace plot used to assess chain mixing and stationarity, while panel (b) shows the full posterior pair plot used to inspect the covariance structure and overall regularity.

Most one-dimensional marginals are unimodal and well-defined, with no discrete multimodality, and the high-probability regions in the two-dimensional panels are connected. The pair plot reveals interpretable coupling: $M_\star$, f_{bulge} , and R_e covary strongly, as expected for parameters that jointly set the inner baryonic support, while the halo parameters trade off against those quantities at the level of the rotation-curve amplitude. The posterior is geometrically regular and suitable as a sampled input to the population model.

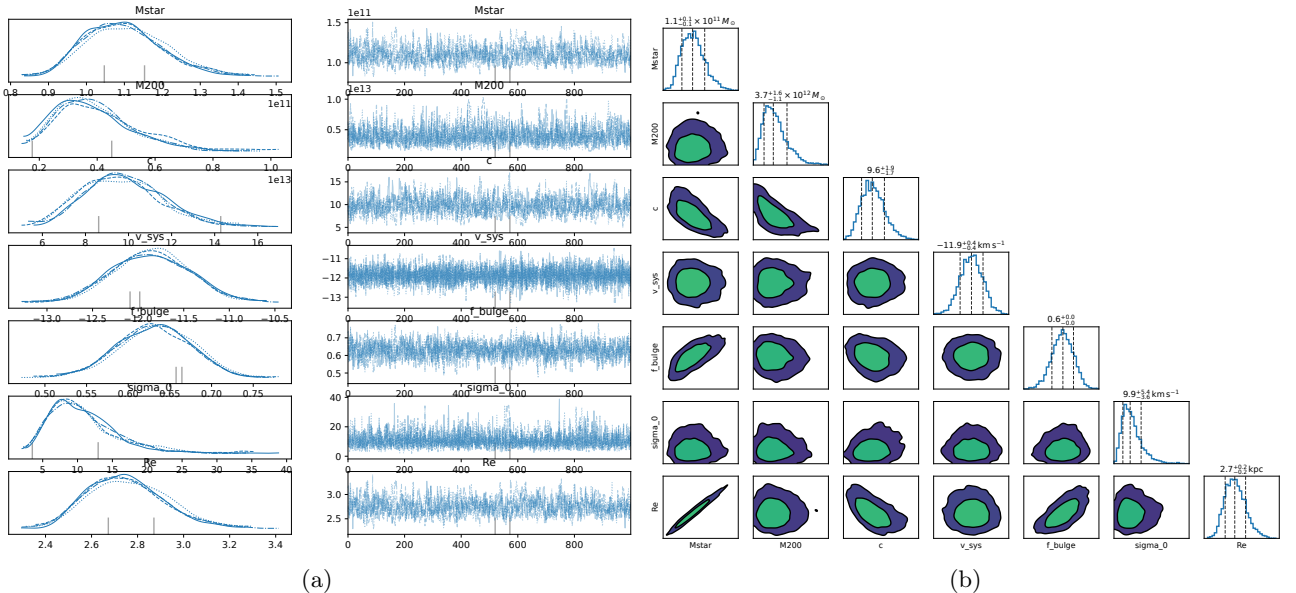


FIG. 8.— Posterior diagnostics for the representative galaxy 11743-9102. *Left*: MCMC trace plot used to inspect chain mixing, stationarity, and mode separation. *Right*: Full posterior pair plot used to assess unimodality, connectedness, and parameter coupling in the single-galaxy posterior.

RESIDUALS OF THE POPULATION C - M FIT

Figure 9 shows the residual-space diagnostics for the population fit on the full sample of 620 galaxies.

The left panel of Figure 9 shows the distribution of $\Delta\log_{10} c \equiv \log_{10} c_{\text{obs}} - \log_{10} c_{\text{pred}}$ as a function of M_{200} . The

horizontal black line denotes zero residual, the blue shaded band denotes $\pm\sigma_{\text{int}}$, and the wider light-gray shaded band denotes $\pm 2\sigma_{\text{int}}$. The figure shows that most points scatter around zero without displaying a systematic monotonic drift with halo mass. This indicates that the sample as a whole does not exhibit a strong unmodeled mass-dependent residual trend, and that the fitted negative slope together with the intrinsic-scatter term provides a reasonably consistent summary of the retained sample.

The right panel of Figure 9 shows the distribution of the standardized residuals $\Delta \log_{10} c / \sigma_{\text{int}}$, overlaid with a unit-Gaussian reference curve. In the current sample, 60.5% of the objects satisfy $|\Delta \log_{10} c| / \sigma_{\text{int}} \leq 1$, and 85.5% satisfy $|\Delta \log_{10} c| / \sigma_{\text{int}} \leq 2$. Both fractions fall below the Gaussian expectations of 68.3% and 95.4%, which is consistent with the factored Student- t population model adopted here: the inferred degrees-of-freedom parameter ν allows for heavier tails than a normal distribution, placing more weight in the wings of the residual distribution. A small number of long-tail outliers deviate more strongly from the median relation.

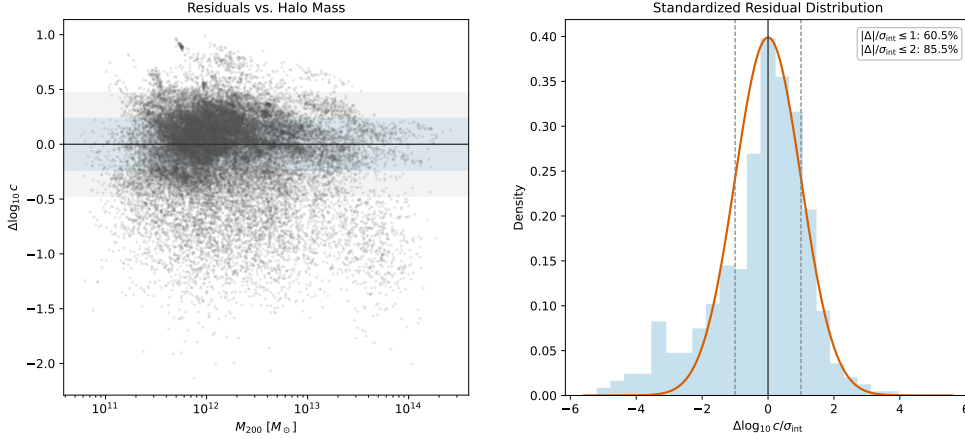


FIG. 9.— Residuals of the population c - M fit for the full sample of 620 galaxies. *Left*: $\Delta \log_{10} c \equiv \log_{10} c_{\text{obs}} - \log_{10} c_{\text{pred}}$ as a function of M_{200} , with $\pm\sigma_{\text{int}}$ (blue) and $\pm 2\sigma_{\text{int}}$ (light gray) bands. *Right*: Distribution of standardized residuals $\Delta \log_{10} c / \sigma_{\text{int}}$ overlaid with a unit-Gaussian reference.

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